

Week 3

Limits of functions pt 2

Examples

$$\textcircled{1} \lim_{x \rightarrow 3} (x^2 - 2x) = 9 - 6 = 3$$

$$\textcircled{2} \lim_{x \rightarrow a} \frac{x - a}{\frac{1}{x} - \frac{1}{a}}$$

$$= \lim_{x \rightarrow a} \frac{x - a}{\frac{a - x}{ax}}$$

$$= \lim_{x \rightarrow a} (x - a) \cdot \frac{ax}{a - x} = \lim_{x \rightarrow a} \cancel{(x - a)} \cdot \frac{ax}{-\cancel{(x - a)}} = \lim_{x \rightarrow a} ax = -a^2$$

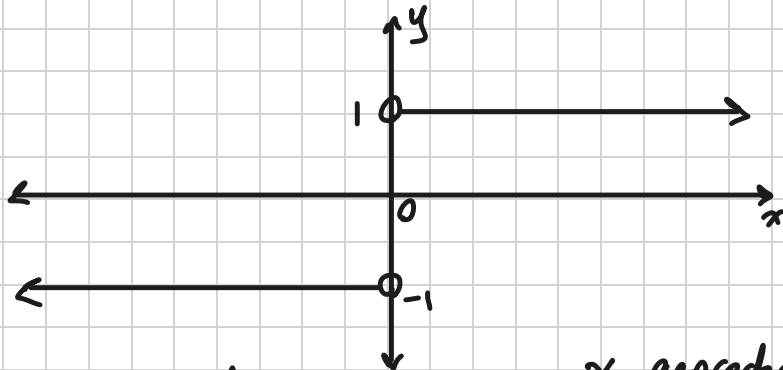
$$\begin{aligned}
 & \textcircled{3} \quad \lim_{t \rightarrow 0} \frac{t}{\sqrt{4+t} - \sqrt{4-t}} \\
 &= \lim_{t \rightarrow 0} \frac{t}{\sqrt{4+t} - \sqrt{4-t}} \cdot \frac{\sqrt{4+t} + \sqrt{4-t}}{\sqrt{4+t} + \sqrt{4-t}} \\
 &= \lim_{t \rightarrow 0} \frac{t(\sqrt{4+t} + \sqrt{4-t})}{4+t - (4-t)} \\
 &= \lim_{t \rightarrow 0} \frac{\cancel{t}(\sqrt{4+t} + \sqrt{4-t})}{2\cancel{t}} \\
 &= \lim_{t \rightarrow 0} \frac{\sqrt{4+t} + \sqrt{4-t}}{2} \\
 &= \frac{\sqrt{4+0} + \sqrt{4-0}}{2} \\
 &= 2
 \end{aligned}$$

One sided limits

Ex

$$f(x) = \begin{cases} -1, & x < 0 \\ 1, & x > 0 \end{cases}$$

$$f(x) = \text{sgn}(x) = \frac{1 \times 1}{x} \quad D(f) = \mathbb{R} - \{0\}$$



Left limit $\lim_{x \rightarrow a^-} f(x)$

x approaches a from the left



$$\lim_{x \rightarrow 0^-} f(x) = -1$$

x approaches a
from the right

Right limit $\lim_{x \rightarrow a^+} f(x)$

$$\lim_{x \rightarrow 0^+} f(x) = 1$$

overall
 $\lim_{x \rightarrow 0} f(x)$ D.N.E

theorem for $\lim_{x \rightarrow a} f(x)$ to exist, the limits $\lim_{x \rightarrow a^-} f(x)$, $\lim_{x \rightarrow a^+} f(x)$ must exist and be equal

Example

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{|3x-1| - |3x+1|}{x}$$

$$|x| = \begin{cases} -x, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases}$$

if x is very small, $3x-1 < 0$

$$= \lim_{x \rightarrow 0} \frac{|3x-1| - |3x+1|}{x}$$

$$= \lim_{x \rightarrow 0} \frac{(1-3x) - (3x+1)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{-\cancel{6x}}{\cancel{x}}$$

$$= -6$$

Ex Find $\lim_{x \rightarrow 0^+} \sqrt{x^3 - x}$

① find domain of $f(x) = \sqrt{x^3 - x}$

$\Rightarrow$ require that $x^3 - x \geq 0$

$$x^3 - x = x(x^2 - 1) = x(x-1)(x+1) \geq 0$$

x	$\ominus$	-1	$\ominus$	0	$\oplus$	1	$\ominus$
$x+1$	$\ominus$	0	$\oplus$		$\oplus$		$\oplus$
$x-1$	$\ominus$		$\ominus$		$\ominus$	0	$\oplus$
$x(x^2-1)$	$-$		$+$		$-$		$+$

$$D(f) = [-1, 0] \cup [1, \infty]$$

Small positive values are not in the domain

$$\lim_{x \rightarrow 0^+} \sqrt{x^3 - x} \quad \text{D.N.E}$$

$$\lim_{x \rightarrow 0^-} \sqrt{x^3 - x} = 0$$

this one works
small negative values are in the Domain

$$\lim_{x \rightarrow 0^+} \sqrt[3]{x^3 - x} = 0$$

$$\lim_{x \rightarrow 0^-} \sqrt[3]{x^3 - x} = 0$$

} No domain problems